

Appendix C

Limitations of the Task Definition Proposed by Hoang and Kan (2010)

1. The task begins after important scholarly decisions have already been made

Hoang and Kan (2010) define related work generation as the task of automatically generating the related work section of a scientific paper given a target paper and its corresponding set of cited papers. Although this definition establishes a clear computational objective, it assumes that the writing process begins once the target paper and relevant literature have already been identified.

However, Section 2.1 demonstrates that literature review writing begins considerably earlier. Before writing starts, researchers determine the purpose of the review, identify the relevant debates within the literature, establish the intended contribution of their work, and decide what should be discussed. Appendix A further argues that existing task formulations do not explicitly represent these activities. Consequently, the proposed task captures only the writing stage of related work generation while omitting the preceding scholarly reasoning that determines the content of the review.

2. The task assumes that the relevant literature has already been identified

Although Hoang and Kan (2010) include relevant document identification as one of the three sub-tasks, the overall task definition assumes that the target paper and its cited papers constitute the input to the generation process.

Appendix A argues that this assumption differs from the actual literature review writing process. Human researchers do not begin with the definitive collection of references that will ultimately appear in the manuscript. Instead, they iteratively retrieve, evaluate, compare, and refine the set of relevant studies throughout the literature review process. Consequently, the task definition assumes that one of the most intellectually demanding aspects of literature review writing has already been completed before generation begins.

3. The task emphasises information selection and synthesis rather than the broader literature review writing process

The three sub-tasks proposed by Hoang and Kan (2010) focus on identifying relevant papers, selecting salient aspects of those papers, and generating a topic-biased related work section. This formulation emphasises information selection and synthesis as the primary objectives of related work generation.

However, Section 2.1 characterises literature review writing as a broader scholarly activity involving rhetorical organisation, critical comparison, argument construction, synthesis of evidence, and positioning the contribution of the current research. Appendix A similarly argues that existing approaches are primarily formulated from a natural language processing perspective and focus on improving the generation pipeline rather than modelling the literature review writing process itself. Consequently, the task definition represents only part of the broader scholarly process required to produce an effective literature review.

4. The task does not explicitly represent the planning process

Section 2.1 shows that researchers make several planning decisions before writing begins, including determining what should be discussed, how the review should be organised, and what writing style should be adopted.

Appendix A explicitly argues that these planning activities are not represented in existing task formulations, which instead rely primarily on the textual representation of the target manuscript to guide generation. Although Hoang and Kan incorporate topic-biased generation, topics are used principally to organise generated text rather than to model the scholarly planning process that precedes writing. Consequently, the task definition provides limited representation of the planning activities that shape literature review development.

5. The task does not explicitly recognise that multiple valid related work sections may exist

The proposed task aims to generate the related work section corresponding to a target paper and its associated references. However, Appendix A argues that literature review writing does not have a single correct outcome.

Different researchers may legitimately produce different related work sections from the same body of literature depending on what they choose to discuss, how they organise the review, and whether they adopt a descriptive or integrative writing style. Consequently, the task definition does not explicitly recognise that multiple valid related work sections may exist for the same collection of reference papers, reflecting the inherently interpretative nature of scholarly literature review writing.

6. The task provides limited representation of the communicative and argumentative purposes of literature reviews

Section 2.1 describes literature reviews as serving important communicative purposes beyond describing previous work. These include establishing the research context, identifying research gaps, critically evaluating existing studies, and positioning the contribution of the current research within the broader literature.

By comparison, Hoang and Kan (2010)’s task definition primarily focuses on generating a coherent synthesis of relevant prior work. Although this objective is necessary, it provides limited representation of the broader rhetorical and argumentative functions that charac-

terise literature reviews as scholarly writing. Consequently, the communicative purpose of the literature review is only partially represented within the proposed task definition.

7. The task provides limited support for iterative human interaction

The proposed task is formulated as a sequence of three stages culminating in the automatic generation of a related work section. This formulation implicitly treats generation as a process that progresses from document identification to text production.

However, Appendix A highlights that literature review writing is inherently iterative. Human researchers repeatedly review, revise, reorganise, and refine both the content and structure of a related work section before producing the final manuscript. These revisions often involve reassessing which studies should be discussed, modifying the organisation of the review, and refining the academic argument as the research evolves.

Consequently, the task definition provides limited support for modelling human interaction throughout the iterative development of related work sections. Rather than viewing generation as a collaborative writing process in which researchers continuously influence and refine the evolving review, the task is primarily conceptualised as producing a completed textual output. This limits its ability to represent the interactive nature of scholarly literature review writing that characterises human practice.

8. The task provides limited representation of human accountability and ethical responsibilities

Hoang and Kan (2010) define related work generation primarily as a computational task involving document identification, salient aspect extraction, and text generation. While this formulation establishes the functional components required to generate a related work section, it provides limited consideration of the ethical responsibilities that are inherent in scholarly literature review writing.

Section 2.2.10 and Appendix A emphasise that ethical considerations extend beyond technical performance to include transparency, bias, attribution, academic integrity, responsible use, and, importantly, human accountability for generated content. In academic writing,

authors remain responsible for the accuracy, completeness, interpretation, and appropriate use of cited literature regardless of the tools used to support writing. Consequently, related work generation systems should contribute to preserving and strengthening this human accountability rather than diminishing it.

This perspective is consistent with the concept of AI-in-the-Loop (AI2L) proposed by Natarajan et al. (2025), who argue that AI2L systems are expert-centred. In such systems, human experts retain responsibility for decision-making throughout the entire workflow, while AI provides information, suggestions, or analyses that inform—but do not replace—expert judgement. From this perspective, the objective of related work generation should not be to autonomously produce a final related work section, but to assist researchers throughout the literature review writing process while preserving their accountability for the resulting scholarly work.

Consequently, the task definition primarily characterises related work generation in terms of its computational objectives while providing limited representation of the ethical principle that human authors remain accountable for generated content. A broader conceptualisation of the task should therefore incorporate mechanisms that support human oversight, informed decision-making, and accountability as integral components of the literature review writing process.